Machine Learning - Undergraduate Level Assessment 14 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
   (a) O(1)
   (b) O(N)
   (c) O(log N)
   (d) O(N^2)
2. If your training loss increases with number of epochs, which of the following could be a possible issue with the learning process?
   (a) Regularization is too low and model is overfitting
   (b) Regularization is too high and model is underfitting
   (c) Step size is too large
   (d) Step size is too small
3. Suppose your model is overfitting. Which of the following is NOT a valid way to try and reduce the overfitting?
   (a) Increase the amount of training data.
   (b) Improve the optimisation algorithm being used for error minimisation.
   (c) Decrease the model complexity.
   (d) Reduce the noise in the training data.
4. Which of the following sentence is FALSE regarding regression?
   (a) It relates inputs to outputs.
   (b) It is used for prediction.
   (c) It may be used for interpretation.
   (d) It discovers causal relationships
5. Statement 1| Industrial-scale neural networks are normally trained on CPUs, not GPUs. Statement 2| The ResNet-50 model has over 1 billion parameters.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: Clustering is the best method for predicting the amount of rainfall based on various environmental cues.
7. True or False: The joint probability distribution P(X, Y, Z) is equal to P(X) * P(Y|X) * P(Z|X).
8. True or False: Out-of-distribution detection is often referred to as anomaly detection in machine learning.
9. True or False: A smaller error rate requires fewer test examples to obtain statistically significant results.
10. True or False: The dimensionality of the null space of the matrix A is 2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find average value of the numbers in a given tuple of tuples.
12. Write a function to check whether the given key is present in the dictionary or not.

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